Week 6

Spring 2027 · Section 029 · MW 9:00–10:20 AM · RRC Room 8210

Course Level Objectives:

- Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. (CLO 3)
- Solve linear regression and correlation problems. (CLO 7)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Locate cases with positive and negative residuals on a scatterplot. (CLO 7) **2.** Use technology to find the least-squares regression line. (CLO 7) **3.** Recognize cases when a regression prediction is not appropriate. (CLO 7)	Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 2) Slides: 2.5 Two Quantitative Variables: Linear Regression StatKey – Descriptive Statistics (Two Quantitative Variables)	Participation: Regression Line – wrap-up of 2.5 Your Turn: Putting It All Together (used-car price vs. age: slope, intercept, prediction, residual)	Online Homework: 2.5 Two Quantitative Variables: Linear Regression (due Sun Feb 28, 11:59 PM) Take-Home Quiz 3 (due Wed Feb 24, 11:59 PM) – covers 2.4–2.5 (all objectives) Test 1 (Fri Feb 26 – Sun Feb 28, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Feb 25) (all objectives)	WileyPLUS StatKey Blackboard Respondus
4. List the sample space for a chance process with equally likely outcomes. (CLO 3) **5.** Compute probabilities by counting equally likely outcomes. (CLO 3) **6.** Apply the complement rule. (CLO 3) **7.** Apply the addition rule to find P(A or B) from a two-way table. (CLO 3) **8.** Compute conditional probabilities from a two-way table. (CLO 3) **9.** Apply the multiplication rule for draws with and without replacement. (CLO 3) **10.** Distinguish disjoint events from independent events. (CLO 3)	e-Text: P.1 Probability Rules Interactive Lecture: P.1 Probability Rules Slides: P.1 Probability Rules	Participation: Probability – Rock and Roll Hall of Fame two-way table; cards with and without replacement Your Turn: Probabilities from a Table (job by enrollment status)	Online Homework: P.1 Probability Rules (due Sun Feb 28, 11:59 PM) Take-Home Quiz 4 (due Mon Mar 8, 11:59 PM) – covers P.1, 3.1–3.2 (all objectives) Test 2 (Fri Apr 2 – Sun Apr 4, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Apr 1) (all objectives)	WileyPLUS StatKey Blackboard Respondus